

University of Minho
School of Engineering

Proteomic Profiling of a Syntrophic Co-culture Grown on Oleate and Stearate

Bioinformatics and Systems Biology
Practical Assignment 1

Group 2

Joana Junqueira (PG63499)
Luís Carmo (PG61440)
Miguel Cruz (PG61442)
Matilde Monteiro (PG63508)

Summary

This study compares the proteins of a syntrophic co-culture of *Syntrophomonas zehnderi* and *Methanobacterium formicicum* grown on oleate and on stearate. The proteins were identified and quantified from mass spectrometry data, then annotated and mapped onto KEGG pathways. The oleate sample gave more identifications than the stearate one (46 proteins against 26), with stronger evidence, since 31 of them were Confident against only one in stearate. In both samples, the methanogenesis proteins of *M. formicicum* were the most abundant, whereas the β-oxidation enzymes of *S. zehnderi* were quantified only with oleate. These results agree with the expected syntrophic partnership. However, since each substrate was analysed only once and the stearate identifications are weaker, it is not possible to say whether the differences between them are real.

1. Methods

The proteins of each sample were identified from mass spectrometry data, quantified by spectrum counting, and then annotated and mapped onto metabolic pathways. The sections below describe these steps in order, from the sequence database to the pathway analysis. Every step was applied in the same way to both samples, so that they could be compared.

1.1. Samples and sequence database

The samples come from a syntrophic co-culture of *Syntrophomonas zehnderi* and *Methanobacterium formicicum* grown on two substrates, oleate (sample 1) and stearate (sample 2).

Each sample consists of 24 MGF files, corresponding to 24 fractions of an SDS-PAGE gel analysed by tandem mass spectrometry, and was analysed separately, with all of its files searched together in a single run. The search database was built from the UniProt protein sequences of both microorganisms (taxa 690567 and 1204725 [1]) and from the common contaminants of the cRAP database. The two FASTA files were merged with Concatenate datasets [2], and FastaCLI, from the SearchGUI suite [3] [4] [5], then added a reversed decoy sequence for each protein, so that the false discovery rate could be estimated later.

1.2. Peptide identification and validation

The workflow was run in Galaxy [6] and is summarised in Figure 1. As a first step, the search parameters were defined once with the Identification Parameters tool of the same suite and reused for both samples. The peptide search was then done with SearchGUI [4] [7], which runs X!Tandem, MS-GF+ and MyriMatch side by side, since combining several engines improves the number and the reliability of identifications.

Each protein in the database was then cut *in silico* with trypsin, allowing up to two missed cleavages and keeping peptides of 8 to 30 amino acids. Spectra were matched to these peptides with a tolerance of 10 ppm for the precursor masses and 0.5 Da for the fragments, considering charges from 2+ to 4+. Carbamidomethylation of cysteine was set as a fixed modification and oxidation of methionine as a variable one.

For validation, the results of the three engines were combined with PeptideShaker [5]. Using the decoy sequences, it estimates how many matches are expected to be wrong and accepts identifications up to a false discovery rate of 1% at the peptide-spectrum match (PSM), peptide and protein levels. The proteins within this limit are classified as Confident if they also pass the additional quality filters and as Doubtful if they fail one or more of them, whereas proteins outside the limit are classified as Not Validated. PeptideShaker produced an mzIdentML file, a Certificate of Analysis and a Protein Report for each sample, and the last two were used in the following steps.

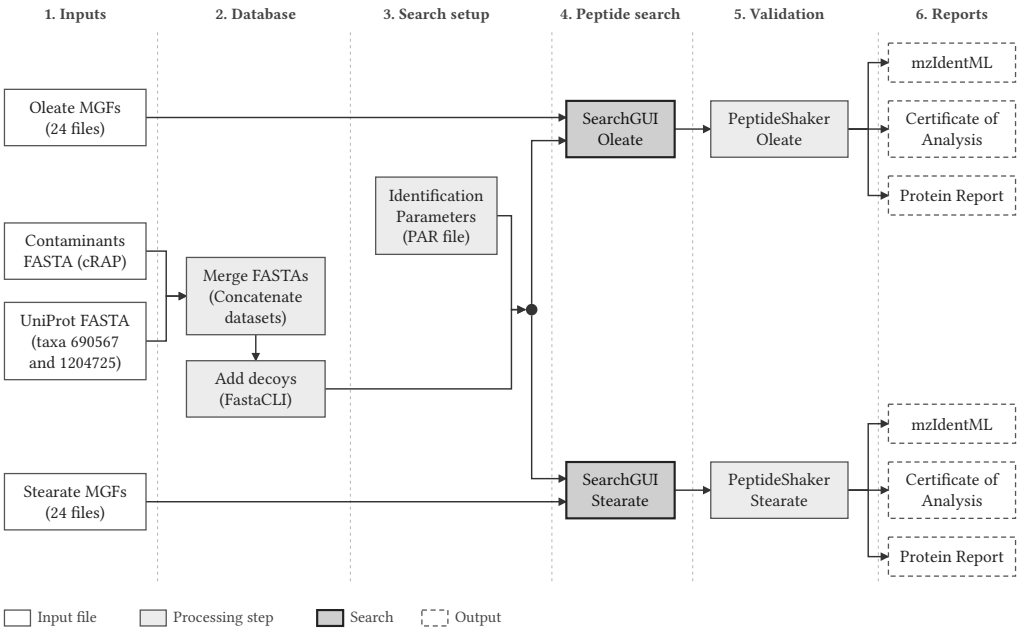


Figure 1: Galaxy workflow for peptide and protein identification in the oleate and stearate samples.

1.3. Relative protein abundance and comparison between samples

The relative abundances were estimated from the Spectrum Counting column of the Protein Report, which PeptideShaker calculates as the normalised spectral abundance factor [8], and the same steps were applied independently to each sample. The contaminant entries (cRAP) were removed and only Confident and Doubtful proteins were kept, so that Not Validated entries were not counted as identifications.

For each sample, the Spectrum Counting values of the remaining proteins, from both microorganisms together, were added up, and the value of each protein was divided by this sum and multiplied by 100. This gives its estimated relative abundance as a percentage of the sample total, with a separate denominator for each sample.

The two samples were combined in a comparison table that lists every protein accession detected in either of them, without duplicates, together with its abundance and validation category in each sample (the proteins selected for the analysis are shown in Figure 2 and Table 1).

A protein not detected in one sample is shown as NA and is not considered absent, since it may not have been identified or validated. The proteins that were identified but had no validated peptides have a Spectrum Counting value of zero, and they are shown as 0.00 to keep them apart from the undetected ones.

1.4. Protein annotation and sequences

The protein accessions of each sample, taken from the filtered Protein Reports, were submitted to the UniProt ID Mapping tool [9], using UniProtKB accessions as the source and UniProtKB as the target database. The annotation tables were set to include the protein name, gene name, organism, length, EC number and Gene Ontology terms. The two samples were handled separately, and both the tables and the protein sequences, in FASTA format, were downloaded for each of them.

The proteins were then organised by microorganism, using the organism identifier of each entry (taxon 690567 for *S. zehnderi* and taxon 1204725 for *M. formicicum*), so that the functions of the two organisms could be interpreted separately. The tables were split with a Python script, and the accessions of each

microorganism were submitted to UniProt again to obtain one FASTA file for each organism and sample.

1.5. Assignment of KEGG Orthology identifiers

To place the proteins in metabolic pathways, each one was assigned a KEGG Orthology identifier (K number) with BlastKOALA [10]. The K numbers group proteins that share the same function across organisms, which gives the proteins of both microorganisms a common link to the KEGG pathway maps. The FASTA files were searched against the KEGG prokaryote database, with one job for each organism and sample (four in total), so that the two microorganisms and the two substrates could be kept apart. The output of each job was converted into a table of proteins and K numbers with a Python script. The proteins without a K number were kept in the tables and marked as NA, and they were not used in the pathway analysis.

1.6. Pathway mapping

The K numbers of each microorganism and sample were submitted separately to the KEGG Mapper Reconstruct tool [11], which lists the KEGG pathways that contain them [12]. This gave four lists of pathways, one for each organism and substrate.

The scripts, the input files and the comparison table with all 50 proteins are available in the branch wa1 of the project repository (github.com/luisvc004/uminho-bbs).

2. Results and interpretation

2.1. Identifications

After removing contaminants, 46 proteins were quantified in the oleate sample (36 from *M. formicicum* and 10 from *S. zehnderi*) and 26 in the stearate sample (22 from *M. formicicum* and 4 from *S. zehnderi*), which gives 50 different proteins in total, 22 of them found in both samples. Because there are no replicates, the comparison is descriptive only, and the absence of detection of a protein in one of the samples does not necessarily imply that it is actually absent.

Table 1: Selected enzymes, with their K number, KEGG pathway and the reaction they catalyse.

Gene	Protein name	K number	Reaction catalysed
Methane metabolism (*M. formicicum*)			
mer	5,10-Methylenetetrahydro-methanopterin reductase	K00320	Reduces 5,10-methylene-H_4MPT to 5-methyl-H_4MPT, using coenzyme F_{420}
mcrA, *mcrB*	Methyl-coenzyme M reductase, subunits alpha and beta	K00399, K00401	Methyl-coenzyme M and coenzyme B give methane and the CoM-S-S-CoB heterodisulfide (last step of methanogenesis)
mvhD	Methyl-viologen-reducing hydrogenase subunit delta	K14127	Oxidises H_2 and uses the electrons to reduce the heterodisulfide
frhB, *frhG*	Coenzyme F420 hydrogenase, subunits beta and gamma	K00441, K00443	Oxidises H_2 and reduces coenzyme F_{420}
cdhA	Acetyl-CoA decarbonylase/ synthase complex subunit alpha	K00192	Interconverts CO_2 and CO in the acetyl-CoA pathway
Fatty acid degradation and electron transfer (*S. zehnderi*)			
fadA	Thiolase	K00632	Cleaves 3-ketoacyl-CoA into a shorter acyl-CoA and acetyl-CoA (last step of β-oxidation)
etfA, *etfB*	Electron transfer flavoprotein, subunits alpha and beta	K03522, K03521	Passes electrons from β-oxidation to the electron transport chain
Oxidative phosphorylation (*S. zehnderi*)			
atpA, *atpD*	ATP synthase, subunits alpha and beta	K02111, K02112	Synthesises ATP from ADP and phosphate, driven by the ion gradient

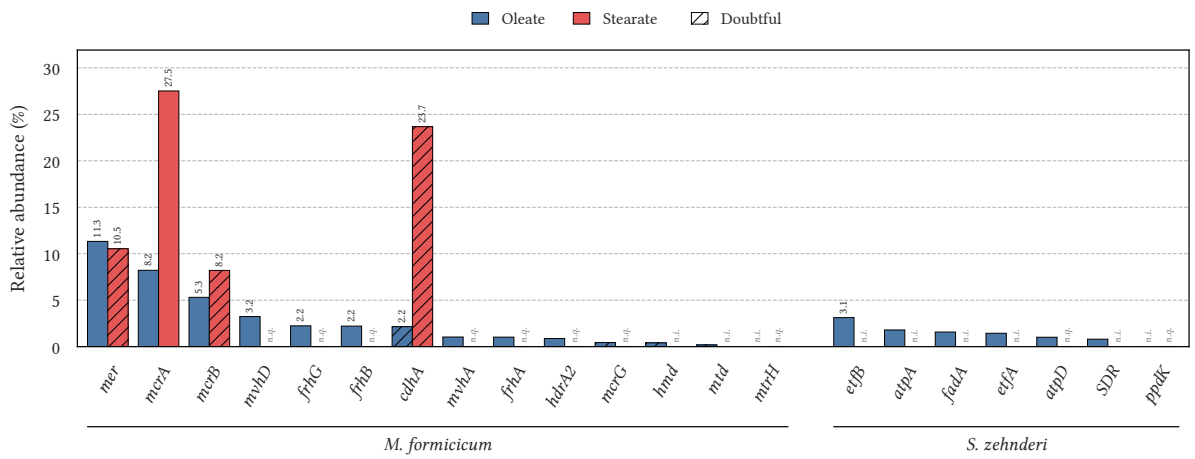


Figure 2: Relative abundance of the proteins selected for the pathway analysis, labelled by the name of their gene, as a percentage of the total of each sample. Hatched bars are Doubtful proteins; n.i. means not identified and n.q. means identified but not quantifiable.

2.2. Methanogenesis in *M. formicicum*

Proteins involved in hydrogenotrophic methanogenesis dominated the protein profiles of both samples (Table 1). The methyl-coenzyme M reductase subunits *mcrA* and *mcrB*, which catalyse the final step of methanogenesis, were among the most abundant proteins detected: *mcrA* represented 8.2% of the oleate profile and 27.5% of the stearate profile, and *mcrB* represented 5.3% and 8.2%, respectively (Figure 2). The protein encoded by *mer*, which acts earlier in the pathway, was also detected in both samples, representing 11.3% in oleate and 10.5% in stearate. The hydrogenase subunits *mvhD*, *frhB* and *frhG* were quantified only in the oleate sample, at 3.2%, 2.2% and 2.2%, respectively, since in stearate they were identified but had no validated peptides. The remaining protein in Table 1, *cdhA*, was detected in both samples, at 2.2% in oleate and 23.7% in stearate, although it is classified as Doubtful in both.

2.3. Fatty acid degradation in *S. zehnderi*

The *S. zehnderi* proteins in Table 1 (*etfB*, *atpA*, *fadA*, *etfA* and *atpD*) were quantified only in oleate, at low abundances (Figure 2). In stearate, they were not identified, except *atpD*, which was identified but not quantifiable. Because of this, it was not possible to compare *S. zehnderi* between the two substrates, and this does not mean that these proteins are absent from the stearate culture.

2.4. Link to the syntrophic co-culture

In *S. zehnderi*, fatty acids are degraded through β-oxidation. During this process, thiolase (*fadA*) releases acetyl-CoA, while the electron-transfer flavoprotein subunits *etfA* and *etfB* transfer electrons to the electron transport chain, which also includes the ATP synthase subunits *atpA* and *atpD*. This process ultimately produces H_2 and acetate.

In *M. formicicum*, H_2 is consumed by the hydrogenases *mvhD*, *frhB* and *frhG*, which reduce the heterodisulfide and coenzyme F_{420}. The reduced F_{420} is then used by *mer* to reduce CO_2-derived carbon to a methyl group, which is transferred to coenzyme M. Finally, methyl-coenzyme M reductase, whose subunits include *mcrA* and *mcrB*, converts methyl-coenzyme M into methane.

By consuming H_2, *M. formicicum* maintains a low hydrogen concentration, thereby making fatty acid oxidation thermodynamically favourable for *S. zehnderi*. The detection of proteins associated with both processes is consistent with the expected syntrophic relationship [13]. However, these observations do not prove that the pathways were complete or actively functioning.

2.5. Oleate versus stearate

Overall, the two samples gave a similar protein profile. In both samples, the methanogenesis proteins of *M. formicicum* accounted for most of the detected protein abundance, whereas proteins from *S. zehnderi* were present at relatively low levels. The main difference involved a small number of *M. formicicum* proteins that were quantified in only one sample. Since oleate and stearate differ only in their degree of saturation and are degraded by the same organisms, a similar profile is not unexpected. However, with no replicates, it is not possible to determine whether the observed differences were caused by the substrate type or by variation between samples.

3. Conclusion and future work

Overall, the results are consistent with the expected syntrophic partnership, since the methanogenesis proteins of *M. formicicum* were found with both substrates and the β-oxidation enzymes of *S. zehnderi* only with oleate. It is harder to say how much of this difference comes from the substrate, because the stearate sample gave fewer and weaker identifications and each substrate was analysed only once. Repeating the experiment with biological replicates would help to separate real differences from ordinary variation, and further studies measuring the acetate, hydrogen and methane produced would show whether these pathways are actually working, since finding an enzyme does not prove that its pathway is active.

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